

Project Re-Ignite Programming Contest 2026 - Mock

A. Party of Hero Alam

4 seconds, 512 megabytes

n number of Hydrogens and n number of Oxygens are going to attend the "Independence Day Chemical Party 2022" today. Hero Alam is the host of the event. But there is a problem. Only n number of chairs are available there in the event. So, it will not be possible to accommodate all of the Hydrogens and Oxygens there. Here, Oxygen is denoted by 'O' and Hydrogen is denoted by 'H'.

No worries, the guests will happily accept that. But there is another universal fact which you must know that Oxygens always like to sit in the middle between two Hydrogens as they always try to form water together. No two Oxygens sit side by side but two Hydrogens may sit side by side. For example: "HOHHOH" is a valid sitting combination for $n = 6$, but "OHHOOH" is not valid, because two Oxygens are sitting side by side here and at the same time the Oxygen at the leftmost chair has no Hydrogen at it's left.

As Hero Alam is the host of the event, now he is trying to generate all possible valid combinations of the seat plan before the arrival of his guests. This can be generated very easily by writing a program but Hero Alam is currently busy with managing the party. So he is requesting you to write a program for it. And also he has a great offer for you. If you can write the program successfully then you are going to be the main character of his next movie. Will you miss this chance?

Input

Only line of input contains a single integer n ($1 \leq n \leq 30$).

Output

Print all possible combinations separated by a single newline in lexicographical order.

input
1
output
H

input
5
output
HHHHH HHHOH HHOHH HOHHH HOHOH

B. Chocolate Factory

3 seconds, 512 megabytes

Rayhan has a chocolate factory. He decided to hire n employees for the production of chocolate in his factory.

In the production line, each employee is positioned one after another. The productivity of each employee depends on the employees near him. The productivity of the i^{th} employee on the k^{th} day is denoted by $P_{i,k}$. The productivity of that employee on the next day is increased by the summation of the productivity of the nearest l_i employees on his left and the nearest r_i employees on his right on the k^{th} day.

If an employee has less than l_i employees on his left, his productivity will depend on all the people available on his left. Similarly, if he has less than r_i people on his right, his productivity will depend on all the people available on his right.

$$P_{i,k+1} = P_{i,k} + \sum_{j=\max(1,i-l_i)}^{i-1} P_{j,k} + \sum_{j=i+1}^{\min(i+r_i,n)} P_{j,k}$$

Now, Rayhan needs to calculate the amount of chocolate produced by each employee (productivity of that employee) on the m^{th} day. Help him solve the problem.

Input

The first line contains two integers— the number of persons working in the factory, n ($1 \leq n \leq 10^5$) and the number of days to consider m ($1 \leq m \leq 100$).

Each of the next n lines contains three integers— P_i , l_i and r_i ($1 \leq i \leq n$).

P_i is the productivity of the i^{th} employee ($0 \leq P_i \leq 10^{12}$).

l_i is the number of employees on the left of the i^{th} employee ($0 \leq l_i \leq 10^5$).

r_i is the number of employees on the right of the i^{th} employee ($0 \leq r_i \leq 10^5$).

Output

Print an array of numbers $P_{i,m}$ ($1 \leq i \leq n$), separated by spaces modulo $10^9 + 7$, the productivity of each employee on m^{th} day.

input
5 3 44 3 0 0 1 2 39 3 1 47 0 2 6 1 1
output
44 864 864 212 212

C. Event Management

1 second, 512 megabytes

MIST wants to develop a system which can calculate the number of students who are capable of an event and this task is assigned to Avash. But Avash is very lazy and he knows that his juniors are really very good at problem solving. So he asked them to solve the problem.

Every student in MIST is marked with two integers a and b . A student is capable of an event, x if $a \leq x \leq b$ or $b \leq x \leq a$. You need to write a program that can calculate the total number of students who are capable of working on an event.

Input

First line of the input contains an integer n ($1 \leq n \leq 10^5$)— the total number of students.

Then the next n lines follow— each line contains two integers, a and b ($1 \leq a, b \leq 10^5$).

The next line contains an integer q ($1 \leq q \leq 10^5$)— the number of events.

The next q lines contain integers x_i ($1 \leq x_i \leq 10^5$)— the value of the i^{th} ($1 \leq i \leq q$) event.

Output

For each event, you have to print the number of students who are capable of working on that event separated by a single newline.

input
5 1 2 2 3 1 4 3 10 7 13 4 3 7 9 14
output
3 2 2 0

D. Palindrome Division

1 second, 512 megabytes

Zarif is fond of palindrome numbers. Palindrome numbers are the number that remains the same when its digits are reversed. For example: 958859 is a palindrome number, where 635 is not. Since Zarif has been very much busy recently, he requests you to help him by doing some tasks.

In fact, you are given a palindrome number n with even number of digits (without leading zeroes). Print another palindrome number with even number of digits that can divide the given number. If there are multiple answers print the minimum one.

Input

Input consists of just a single line, the given number n ($10 < n < 10^9$).

Output

Print a single line containing the answer.

input
44
output
11

E. The Mark of XORO

1 second, 512 megabytes

Problems - Codeforces

For an array arr of n positive integers, count the number of unordered pairs (i, j) ($1 \leq i < j \leq n$) where $arr_i \oplus arr_j > arr_i \& arr_j$. Here, $\oplus$ denotes the bitwise XOR operation and $\&$ denotes the bitwise AND operation.

Input

The first line of input contains a positive integer n ($1 \leq n \leq 10^5$)— the size of the array. The second line contains n space-separated integers $arr_1, arr_2, \dots, arr_n$ ($1 \leq arr_1, arr_2, \dots, arr_n \leq 2^{30}$).

Output

The number of pairs (i, j) where $arr_i \oplus arr_j > arr_i \& arr_j$.

input
4 4 3 5 2
output
4

input
3 1 7 2
output
3

F. Nothing is Constant Other than CHANGE

1 second, 512 megabytes

The new cafeteria of MIST is spacious, right? To make the best use of the spaces, the authority is trying to implement some innovative ideas. One of them is to place a book corner inside the café.

Musfique visits MIST Café regularly. He noticed that the valuable time of the students was being wasted standing in the queue for buying food. So he thought of using his coding skill to solve this problem.

Within a few days he made a vending machine which is very cost effective. The authority set up several vending machines made by Musfique in the café to encourage him. But the problem is that these vending machines can only take money and supply food but not change. That is, if a person pays a 100 Taka note to buy a food that costs 70 Taka, then he will never get back the remaining 30 Taka. So the person will pay a 50 Taka note and a 20 Taka note instead. Another way of payment can be three 20 Taka notes and one 10 Taka note.

Now find the minimum number of notes someone needs in order to pay t Taka.

Note that, the machine can recognize the mentioned notes only— 1 Taka, 2 Taka, 5 Taka, 10 Taka, 20 Taka, 50 Taka, 100 Taka, 500 Taka and 1000 Taka. Neither any card nor any coin is valid.

Input

The only line of the input contains an integer t ($1 \leq t \leq 10^4$).

Output

Output the minimum number of notes needed to pay the cost of the food.

input
145
output
4

input

814

output

7

G. Musfique's Wallet

2 seconds, 512 megabytes

Getting a huge success by making vending machines, Mushfique is now a rich man. He carries a lot of cash in his wallet. He needs to keep track of his money.

Initially Mushfique's wallet is empty. There are q operations. Each operation can be one of the following types:

- $0\ k$: Puts a note valued k in his wallet.
- 1 : Gives away the latest inserted note that the wallet currently contains.
- 2 : Tries to remember the most valued note that the wallet currently contains.

Note that, only the mentioned notes are available— 1 Taka, 2 Taka, 5 Taka, 10 Taka, 20 Taka, 50 Taka, 100 Taka, 500 Taka and 1000 Taka.

Input

The first line contains an integer q ($2 \leq q \leq 2 \cdot 10^5$)— the number of operations.

The next q lines follow, each of them contains one of the operations mentioned in the statement.

Output

For operation of type 0 : Print nothing.

For operation of type 1 : Print the value of the note that Mushfique is giving away.

For operation of type 2 : Print the most valued note that the wallet currently contains.

input

7

0 2

1

0 10

2

0 5

2

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

1

output

2

10

10

5

5

5

5

5

5

5

5

5

5

5

5

5

5

5

5

5

5

5

H. Vowel Ciphers

2 seconds, 512 megabytes

In cryptography, a *Caesar Cipher* is a type of substitution cipher in which each letter in the plaintext is replaced by a cipher alphabet which is the plain alphabet rotated left or right by some number of positions. Note that the rotation is cyclic i.e. the letter after 'z' is 'a' and onwards.

Problems - Codeforces

Sadia was tasked with encoding message strings consisting of lowercase English letters with Caesar Cipher right shift. But she loves vowels. So, she wants to maximize the number of vowels in the message strings. Help her achieve his goal.

You will be given a string s and q queries, each query will be of one of the 3 types:

- $E\ x$: You have to right shift cipher the string s by x positions ($1 \leq x \leq 1000$).
- $R\ i\ c$: You have to replace the character at the i^{th} index with the lowercase letter c ($0 \leq i < |s|$).
- V : You have to print by how much minimum position one right shift cipher needs to be done on string s to get the maximum number of vowels and also print the maximum number of vowels you can get, separated by a space.

In this problem the vowels of the English alphabet are 'a', 'e', 'i', 'o' and 'u'.

Input

The first line of the input contains an integer t ($1 \leq t \leq 10^5$)— the number of test cases.

The descriptions of the test cases follow.

The first line of the description of each test case contains one string s ($1 \leq |s| \leq 10^5$) consisting of lowercase English letters.

The second line contains an integer ($1 \leq q \leq 10^5$)— the number of queries.

The next q lines follow, each of them contains one of the query types mentioned in the statement.

It is guaranteed that the sum of $|s|$ over all test cases does not exceed 10^5 and the sum of q over all test cases does not exceed 10^5 .

Output

For each query of the third type, print the minimum number of positions the right shift has to be done and the maximum number of vowels we can get, with a space in between them.

Don't forget to print a new line after each test case.

input

2

xbfjp

5

V

R 0 v

V

E 4

V

cglrxc

3

V

R 5 v

V

V

V

V

V

V

V

V

V

V

V

V

V

V

V

V

V

output

25 4

25 5

21 5

2 3

3 3

3 3

3 3

3 3

3 3

3 3

3 3

3 3

3 3

3 3

3 3

I. Demon Slayer

1 second, 512 megabytes

Tanjiro is a member of the Demon Slayer Corps whose job is to kill demons. In the epic **Demon vs Human** war, Tanjiro is facing n demons. The i^{th} ($1 \leq i \leq n$) demon has a power of a_i ($a_1, a_2, \dots, a_n$). As a human, Tanjiro also has a health h . After killing the i^{th} demon, his health decreases by a_i . Now, Tanjiro wants to kill as many demons as possible while he is alive. What is the maximum number of demons he can kill before dying as a hero.

Input

The first line of the input contains n ($1 \leq n \leq 10^5$)— the number of demons and h ($0 \leq h \leq 10^9$)— the initial health of Tanjiro. The next line contains n space-separated integers a_i ($1 \leq a_i \leq 10^9$) describing the power of the i^{th} ($1 \leq i \leq n$) demon.

Output

Output the maximum number of demons Tanjiro can kill.

input
5 10 6 4 2 3 5
output
3

input
1 10 11
output
0